

Homework 4 Solutions

Question 1

(50 points) The Study of Assets and Health Dynamics among the Oldest Old (AHEAD) is a national longitudinal study with initial sample of 7444 respondents aged 70 years and older, and their spouses (if married). Objectives of the study include: (i) to monitor transitions in physical, functional, and cognitive health; (ii) to examine the relationship of late-life changes in physical and cognitive health to patterns of saving and income flows; (iii) to relate changes in health to economic resources and intergenerational transfers; (iv) to examine how the mix and distribution of economic, family, and program resources affect key outcomes, including institutionalization, saving, and health declines. In this problem, we look at data from the first four waves of the study, collected in 1993, 1995, 1998 and 2000. A reduced data set ahead.xlsx is on the course website. We explore the relationship of a test of cognitive function to some physical functioning indicators. The hypothesis is that, as physical function declines, movement becomes more difficult, and hence there are fewer stimuli, leading to cognitive decline. Here is a description of the variables contained:

```
id- subject identification number
year- study wave
sex- 1=male,2=female
age- age in years
immword- immediate word recall
delword- delayed word recall
blks- difficulty walking several blocks without help
strs- difficulty climbing flight of stairs without help
push- difficulty pulling or pushing a living room chair
bag- difficulty lifting a bag of groceries
dime- difficulty picking up a dime
iadlany- perform any instrumental activities of daily living
```

The focus of the study is to assess the association of difficulty performing instrumental activities of daily living (IADLs) to memory and cognition problems. IADLs include activities such as shopping, managing money, taking medications, using the telephone, preparing meals. They are supposed to represent the types of integrative activities that one needs to be able to perform in order to live independently in the community.

Here, we'll use total word recall ($\text{totword} = \text{immword} + \text{delword}$) as our measure of cognitive decline and iadlany , an indicator for whether or not the subject reports difficulty with any (at least one) IADL, as our measure of physical function decline. The goal of this analysis is to determine if there is a relationship between total word recall and the IADL variable using GEE. Consider the variables age and sex as possible confounders in this analysis. You do not have to consider any effect modifiers.

(a) (5 points) Discuss whether you feel that GEE is the appropriate analysis choice for this project.

Either answer is acceptable with appropriate justification. It really depends on what you are interested in.

For GEE: GEE is a good choice because total word recall is not a modifiable risk factor. Here, we are interested in comparing the population with IADL to the population without IADL. This will give us an idea of how these populations are different.

For GLMM: GLMM is a good choice because we are interested in the increased risks a person will have as their total word recall changes. This will tell us what to expect in terms any increased risk for decreased total word recall.

If you used total word recall as the outcome:

For GEE: GEE is a good choice because IADL is not a modifiable risk factor. Here, we are interested in comparing the population with IADL to the population without IADL. This will give us an idea of how these populations are different.

For GLMM: GLMM is a good choice because we are interested in the increased risks a person will have as they transition someone without IADL to someone with IADL. This will tell us what to expect in terms any increased risk for decreased total word recall.

(b) (45 points) Complete an analysis of this data using a GEE that results in the best possible answer to the scientific aims of the study. This should start with exploratory analysis of the data, explore different statistical models based on model fit and scientific hypotheses, and finish with a clear summarization of your findings including the interpretation of the coefficients of interest. Please include only relevant output to how you found the best model. It is encouraged to give the code to replicate your findings.

Which outcome?

In the past, some have questioned using iadlany as the outcome variable (previous version of the question was more clear that it should be used as the outcome). Many felt that total word recall was the more natural outcome variable. It does state in the question *“The hypothesis is that, as physical function declines, movement becomes more difficult, and hence there are fewer stimuli, leading to cognitive decline”* which sounds like iadlany impacts cognitive decline (i.e., total word recall).

Overall, I agree that total word recall is the more nature outcome. However, I have always thought of both of these analyses as being associations versus causal relationships, so the solutions for both analyses are included below and no credit will be taken off for either approach.

What is being asked?

Note that this question is not necessarily a “longitudinal” question (like a required for the midterm project). Let’s look at the goal of the study:

“The goal of this analysis is to determine if there is a relationship between total word recall and the IADL variable using GEE.”

This can be answer from cross-sectional data since all were interested in is the relationship between IADL and total word recall. It doesn’t necessarily need an interaction term between IADL and time/age. It *can* have an interaction term but that isn’t required. All we need to answer this question is to have IADL in the model.

Solution with iadlany as the outcome:

The iadlany variable is a 0/1, so here we’ll be using logistic regression. The main challenge with logistic regression is to determine how to include continuous variables. So, the first thing I’m going to explore how iadlany depends on age of the subject and on total word recall.

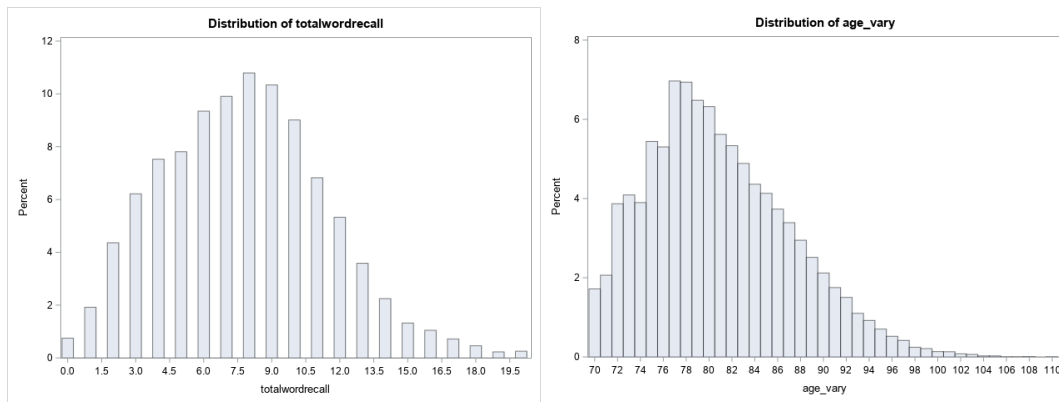
Since this is a large dataset, I’m going to explore the relationship between the log odds of iadlany with age and total word recall. To do this I will:

1. get a table of the probabilities of iadlany by the word recall variable,
2. transform the probabilities to the logit scale (since that's what we'll model things on), and
3. plot the logit probabilities by total word recall

Then we’ll do the same for age (code not shown). For age, I’m going to create a time-varying age variable (the data only gives baseline age). We could, explore the impact of baseline age and time in study separately, but here I’m only going to look at age as a time varying covariate.

First I’m going to look at the distribution of the variables. This is important to know where the data are. So that the figures can be interpreted appropriately. If only 1 percent of the observations are at a particular age or total word recall, then I don’t want to over interpret the trends that I see at that point. **I’m not looking at these figures to see if the variables are normally distributed. Since these variables are not the outcome, I don’t care if they are normally distributed.**

```
proc univariate data=BIOS755.ahead2;  
  histogram totalwordrecall age_vary;  
run;  
quit;
```



So, in the figures we should really focus on total word recall for the range of 2 – 15 or 16, and for age we should really focus on < 95 or so.

Usually I would estimate the average using a lowess smoother (I do this below when total word recall is the outcome). Here, I calculate the mean for each value of age and total word call and then plot the logit transformation by those variables. This works fine here, since we have a large sample size. Overall, I think the lowess smoother is a better option for most scenarios.

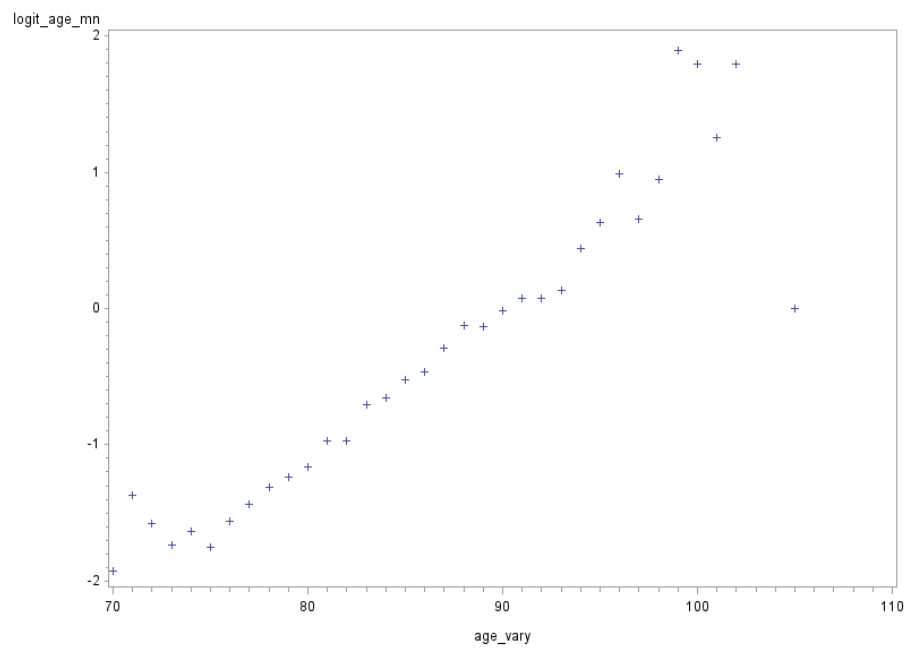
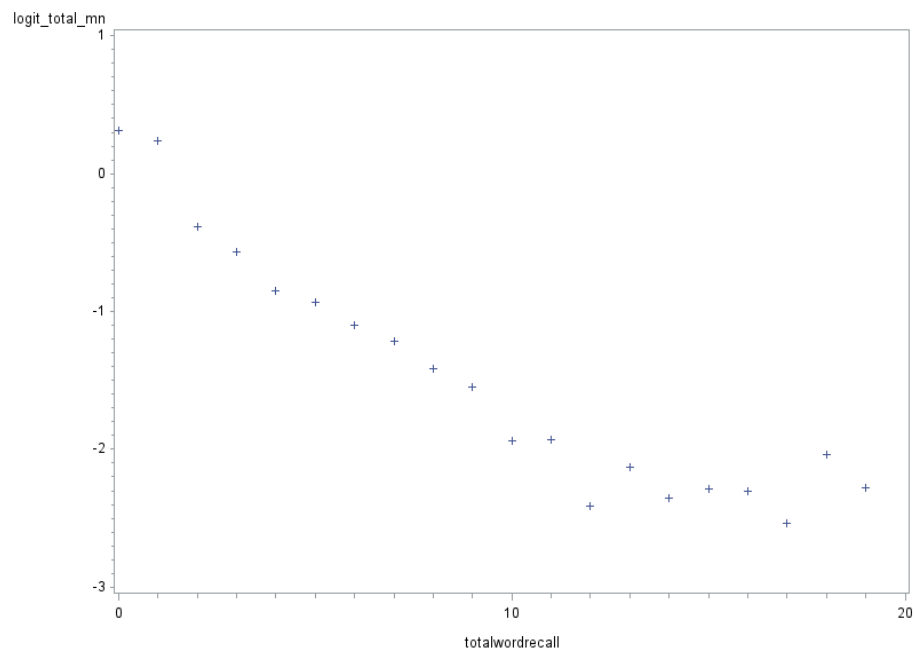
```
data ahead2;
set work.ahead;
totalwordrecall = immword + delword;
age_vary = age + year;
run;

proc sort data=ahead2;
by totalwordrecall;
run;

proc means mean data=ahead2;
by totalwordrecall;
var iadlany;
output out=total_mn mean=iad_total;
run;

data logit_total;
set total_mn;
logit_total_mn = log(iad_total/(1-iad_total));
run;

proc gplot data=logit_total;
plot totalwordrecall*logit_total_mn;
run;
quit;
```



Both of these plots suggest a linear relationship between them and the outcome (on the logit scale) seems satisfactory. For total word recall, possibly a log-linear or quadratic term may be good to add. Here, I'll test out a quadratic term.

```
proc sort data=ahead2;
by id;
run;
```

```

proc gee data=ahead2 DESCENDING;
class id sex;
model iadlany = age_vary totalwordrecall
totalwordrecall*totalwordrecall sex/link=logit dist=bin;
repeated subject=id/corr=exch corrw;
run;

```

The SAS System

The GEE Procedure

Model Information

Data Set	WORK.AHEAD2	iadlany
Distribution	Binomial	
Link Function	Logit	
Dependent Variable	iadlany	

Number of Observations Read	26604
Number of Observations Used	18907
Number of Events	4134
Number of Trials	18907
Number of Missing Values	7697

Class Level Information

Class Levels Values

id	6651	1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33 34 35 36 37 38 39 40 41 42 43 44 45 46 47 48 49 50 51 52 53 54 55 56 57 58 59 60 61 62 63 64 65 66 67 68 69 70 71 72 73 74 75 76 77 78 79 80 81 82 83 84 85 86 87 ...
sex	2	1 2

Response Profile

Ordered Value	iadlany	Total Frequency
1	1	5686
2	0	15476

PROC GEE is modeling the probability that iadlany='1'.

GEE Model Information

Correlation Structure	Exchangeable
Subject Effect	id (6651 levels)
Number of Clusters	6651
Clusters With Missing Values	3655
Correlation Matrix Dimension	4
Maximum Cluster Size	4
Minimum Cluster Size	0

Working Correlation Matrix

	Obs 1	Obs 2	Obs 3	Obs 4
Obs 1	1.0000	0.2977	0.2977	0.2977
Obs 2	0.2977	1.0000	0.2977	0.2977
Obs 3	0.2977	0.2977	1.0000	0.2977
Obs 4	0.2977	0.2977	0.2977	1.0000

Exchangeable Working Correlation

Correlation	0.2977
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GEE Fit Criteria

QIC 18346.6925

QICu 18343.5995

Parameter Estimates for Response Model with Empirical Standard Error Estimates

Parameter	Estimate	Standard Error	95% Confidence Limits		Z	Pr > Z
Intercept	-4.6368	0.3193	-5.2626	-4.0110	-14.52	<.0001
age_vary	0.0585	0.0037	0.0512	0.0658	15.66	<.0001
totalwordrecall	-0.1952	0.0166	-0.2278	-0.1626	-11.73	<.0001

**Parameter Estimates for Response Model
with Empirical Standard Error Estimates**

Parameter		Estimate	Standard Error	95% Confidence Limits		Z	Pr > Z
totalword*totalwordr		0.0050	0.0010	0.0030	0.0069	4.93	<.0001
sex	1	-0.4185	0.0488	-0.5141	-0.3230	-8.58	<.0001
sex	2	0.0000	0.0000	0.0000	0.0000	.	.

Note here that we're modeling the probability that iadlany = 1 not 0. So we're modeling the probability that they do have problems performing instrumental activities of daily living. Or, we can interpret things the opposite way by multiplying all the coefficients by -1.

Final verdict:

In this model we have totalwordrecall and totalwordrecall². So, to give the effect of totalwordrecall we need to specify which value we're talking about.

The coefficient at totalwordrecall = 0 is the main effect = -0.1952.

The coefficient at totalwordrecall = 10 is

$$-0.1952 + 10 \times 0.0050 = -0.1452$$

And the coefficient at totalwordrecall = 20 is

$$-0.1952 + 20 \times 0.0050 = -0.0952$$

As predicted from the plot the impact of total word recall on iadlany is decreasing as total word recall gets larger. Let's interpret the coefficient for totalwordrecall = 10:

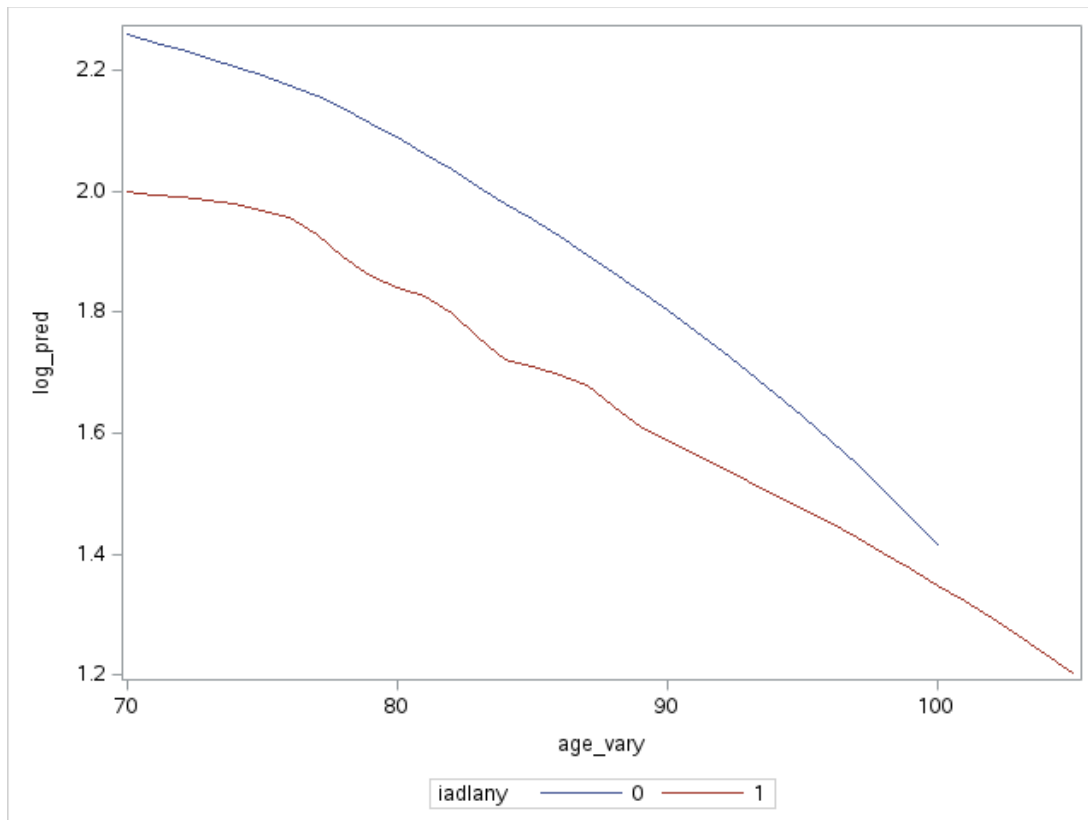
Those with a total word recall of 10 have odds of performing instrumental activities of daily living that are 13.5% lower ($= 1 - \exp(-0.1452)$) than those with a total word recall of 11, after adjusting for age and sex.

According to our model, it appears that there is sufficient evidence to show that total word recall has a significant effect on the probability of having trouble with any IADL tasks. This relationship is diminishing as total word recall is increasing. Further, according to the model, it appears that age and gender have a significant effect on the probability of a person having trouble with any IADL tasks. However, it does not appear that sex changes the effect of total word recall on the probability that a person will have trouble with IADL skills.

Analysis using total word recall as the outcome:

Since total word recall is a count variable, we'll consider the Poisson and Negative Binomial models. Both use a log link. Below, I'll take a look at the relationship between age and the log of average total word recall by the iadlany variable. The code to reproduce this output can be found in the .sas file.

Here, 'log_pred' is the log of the average produced by a lowess smoother.



The pattern appears to be quadratic, not clear if iadlany has any impact.

```
proc gee data=ahead2;  
class id sex (param=REF);  
model totalwordrecall = sex age_vary iadlany age_vary*age_vary/link=log dist=negbin;  
repeated subject=id/corr=exch corrw;  
run;
```

The GEE Procedure

Model Information

Data Set

WORK.AHEAD2

Model Information	
Distribution	Negative Binomial
Link Function	Log
Dependent Variable	totalwordrecall
Number of Observations Read	26604
Number of Observations Used	18907
Number of Missing Values	7697

Class Level Information		
Class	Value	Design Variables
sex	1	1
	2	0

GEE Model Information	
Correlation Structure	Exchangeable
Subject Effect	id (6651 levels)
Number of Clusters	6651
Clusters With Missing Values	3655
Correlation Matrix Dimension	4
Maximum Cluster Size	4
Minimum Cluster Size	0

Working Correlation Matrix				
	Obs 1	Obs 2	Obs 3	Obs 4
Obs 1	1.0000	0.4449	0.4449	0.4449
Obs 2	0.4449	1.0000	0.4449	0.4449
Obs 3	0.4449	0.4449	1.0000	0.4449
Obs 4	0.4449	0.4449	0.4449	1.0000

Exchangeable Working Correlation

Correlation	0.4449
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GEE Fit Criteria	
QIC	-306789.1010
QICu	-306794.1152

Parameter Estimates for Response Model							
with Empirical Standard Error Estimates							
Parameter		Estimate	Standard Error	95% Confidence Limits		Z	Pr > Z
Intercept		-0.5051	0.5841	-1.6499	0.6397	-0.86	0.3872
sex	1	-0.1035	0.0098	-0.1227	-0.0843	-10.57	<.0001
age_vary		0.0883	0.0147	0.0595	0.1171	6.01	<.0001
iadlany		-0.1576	0.0090	-0.1751	-0.1400	-17.56	<.0001
age_vary*age_vary		-0.0007	0.0001	-0.0009	-0.0005	-7.59	<.0001

Here, we can see that iadlany clearly has an impact. $1 - \exp(-0.1576) = 0.146$ or about a 15% decrease. As a result, we can interpret the coefficient as follows:

We estimate that the average rate of total word recall is 15% lower for those with iadlany=1 versus 0, after controlling for age and sex.

Question 2

a) (20 points) Consider a logistic-generalized linear mixed model with random intercepts for the subject-specific log odds of skin cancer with time, treatment, and a time by treatment interaction. Find the best way to include time and the time by treatment interaction.

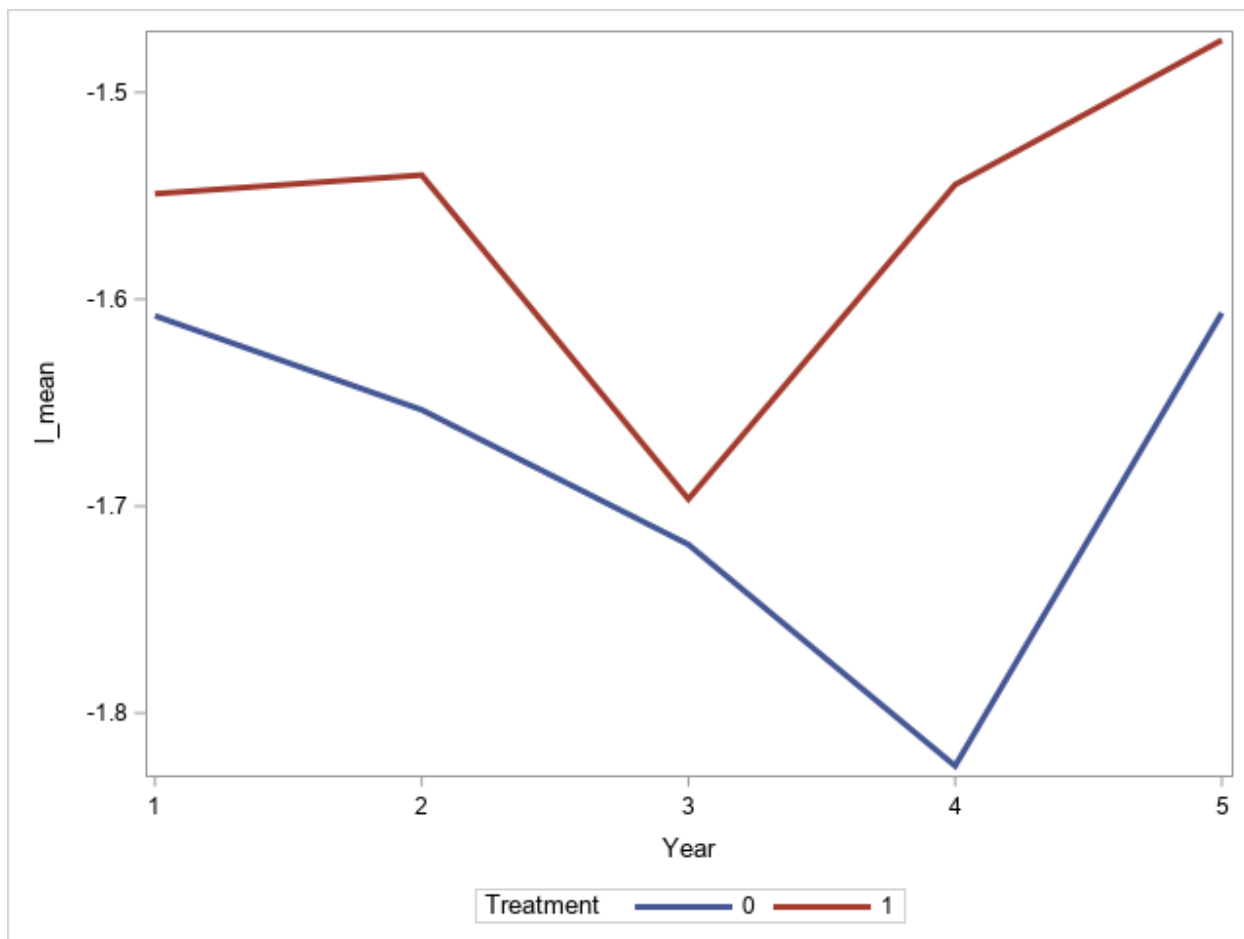
The best way to determine how to include time and the time*treatment interaction is to look at the plot of the logit transformed means over time. The code below creates this plot.

```
proc sort data=skin;
by Treatment year;
run;

*Calculate the mean by week;
proc means mean data=skin noprint;
by Treatment year;
var Y_bin;
output out = MN_TRT_dat mean = mn_TRT_Y_bin;
run;

*transform the means to the logit scale;
data MN_TRT_dat;
set MN_TRT_dat;
l_mean = log(mn_TRT_Y_bin/(1-mn_TRT_Y_bin));
run;

*Let's look at the mean (on the logit scale) by TRT group;
Proc SGplot data = MN_TRT_dat;
series x=year y=l_mean / group =Treatment LineAttrs= (pattern=1 thickness=3);
run;
```



Here's really not much here to go by. Since we only have 5 years and a lot of data, I think profile analysis is the way I would go.

With the binary outcome and a logistic model (with dummy coding for time), the parameter estimates from a model with 20 quadrature nodes are:

Solutions for Fixed Effects							
Effect	Treatment	Year	Estimate	Standard Error	DF	t Value	Pr > t
Intercept			-2.2494	0.1314	1681	-17.11	<.0001
Year		2	-0.05986	0.1539	5390	-0.39	0.6974
Year		3	-0.1246	0.1571	5390	-0.79	0.4278
Year		4	-0.2520	0.1653	5390	-1.52	0.1274

Solutions for Fixed Effects							
Effect	Treatment	Year	Estimate	Standard Error	DF	t Value	Pr > t
Year		5	0.07821	0.1889	5390	0.41	0.6788
Year		1	0	.	.	.	.
Treatment	1		0.08129	0.1707	5390	0.48	0.6340
Treatment	0		0	.	.	.	.
Treatment*Year	1	2	0.08503	0.2131	5390	0.40	0.6899
Treatment*Year	1	3	-0.04694	0.2192	5390	-0.21	0.8304
Treatment*Year	1	4	0.2912	0.2273	5390	1.28	0.2003
Treatment*Year	1	5	0.08677	0.2651	5390	0.33	0.7434
Treatment*Year	1	1	0	.	.	.	.
Treatment*Year	0	2	0	.	.	.	.
Treatment*Year	0	3	0	.	.	.	.
Treatment*Year	0	4	0	.	.	.	.
Treatment*Year	0	5	0	.	.	.	.
Treatment*Year	0	1	0	.	.	.	.

Type III Tests of Fixed Effects				
Effect	Num DF	Den DF	F Value	Pr > F
Year	4	5390	1.26	0.2828
Treatment	1	5390	2.04	0.1537
Treatment*Year	4	5390	0.61	0.6565

It doesn't appear that there's much there for year or treatment.

(b) (10 points) Regardless of significance, give an interpretation for time by treatment interaction.

The treatment-by-year interaction terms indicate that the odds ratios comparing beta-carotene to placebo vary by year, but there is no consistent pattern in the treatment effect over time relative to Year 1.

- **Year 2 (Estimate = 0.0850):**
For a given subject, in Year 2, the odds of having at least one new skin cancer in the beta-carotene group are $\exp(0.0850) = 1.09$ times the odds in the placebo group, relative to Year 1.
- **Year 3 (Estimate = -0.0469):**
For a given subject, in Year 3, the odds of having at least one new skin cancer in the beta-carotene group are $\exp(-0.0469) = 0.95$ times the odds in the placebo group, relative to Year 1.
- **Year 4 (Estimate = 0.2912):**
For a given subject, in Year 4, the odds of having at least one new skin cancer in the beta-carotene group are $\exp(0.2912) = 1.34$ times the odds in the placebo group, relative to Year 1.
- **Year 5 (Estimate = 0.0868):**
For a given subject, in Year 5, the odds of having at least one new skin cancer in the beta-carotene group are $\exp(0.0868) = 1.09$ times the odds in the placebo group, relative to Year 1.

Here's the interpretation if you used quadratic for year (which was OK).

Interaction with year = 0.05142

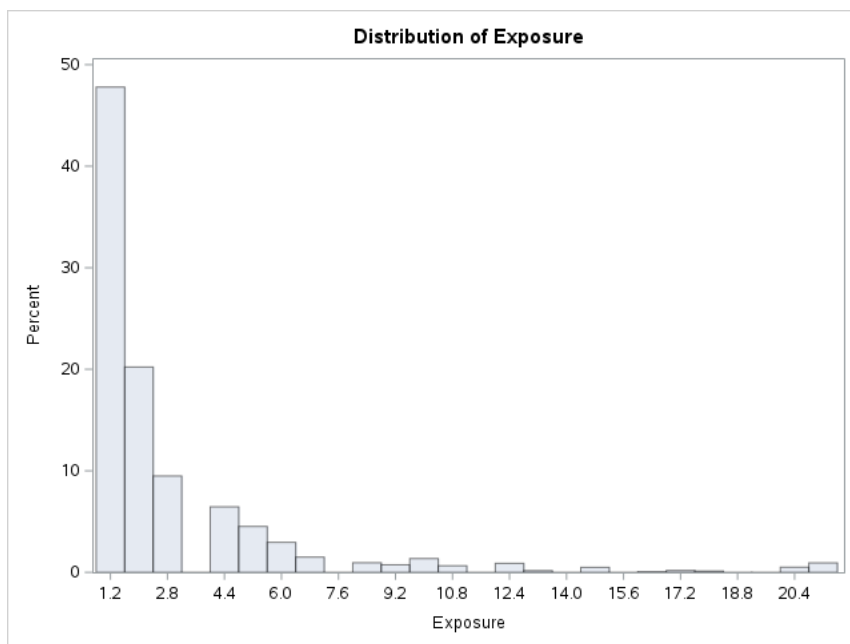
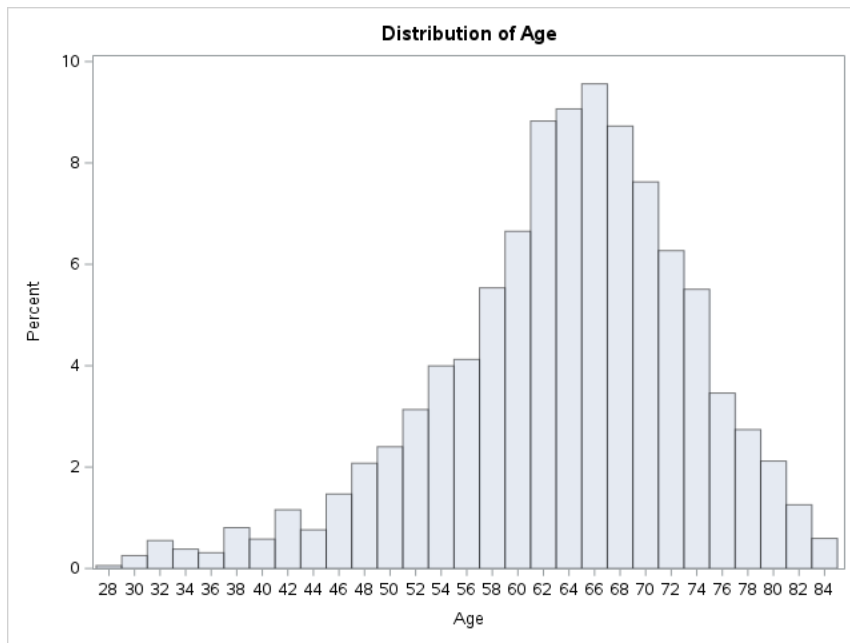
Interaction with year*year = -0.0014

To interpret the interaction, we have to say what year we are referring to.

- **Interpretation of interaction at year 1:** For each additional year at year 1, a subject in the treatment group has their odds of skin cancer increased by a factor of 1.051 ($\exp(0.05142 - 0.0014) = 1.051$, or by 5.1%) versus the same subject in the control group.
- **Interpretation of interaction at year 5:** For each additional year at year 5, a subject in the treatment group has their odds of skin cancer increased by a factor of 1.0454 ($\exp(0.05142 - 5 * 0.0014) = 1.0454$, or by 4.5%) versus the same subject in the control group.

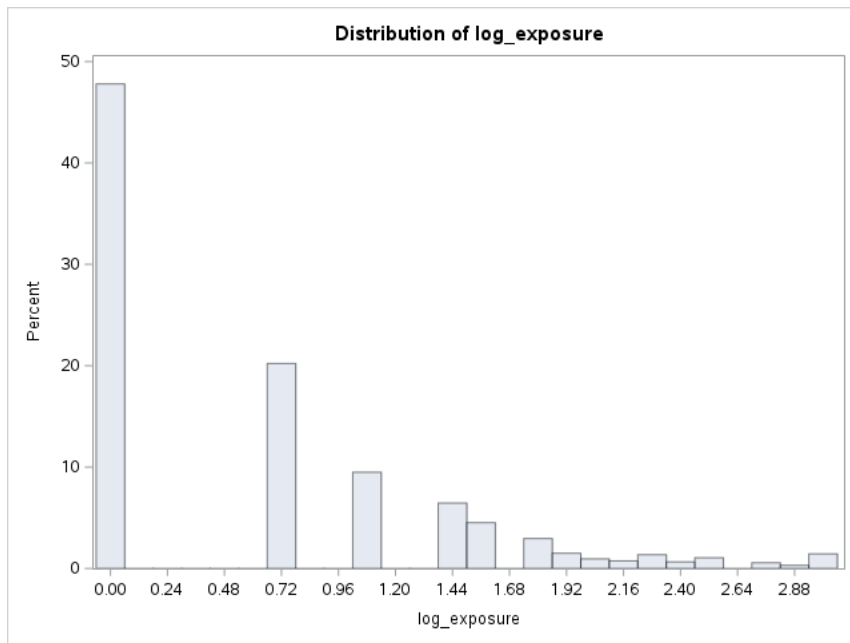
(c) (17 points) Repeat the random effect model analysis in (a) while appropriately adjusting for the exposure and age variables. Give an interpretation for time by treatment interaction.

To start, I'm going to look at a histogram of age and exposure to get an idea of what type of variables these are:



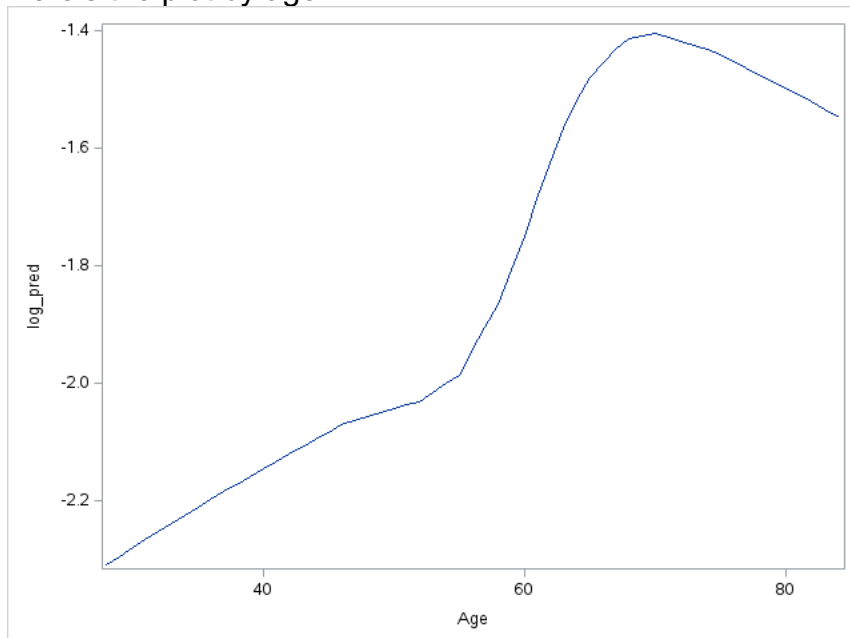
Age has a relatively normal distribution (a little left skewed)

Exposure is pretty right skewed. For these types of variables, using a log transform is commonly a good idea. Here's what the log-transformed version of exposure looks like:



Now, I'll look at the relationship between the outcome and all three of these variables. For each I use the same type of code to get the plots: i) lowess smoother, ii) calculate $\log(\text{pred}/(1-\text{pred}))$ of the lowess predicted (pred) values, and iii) plot the variables. See the sas code for details:

Here's the plot by age

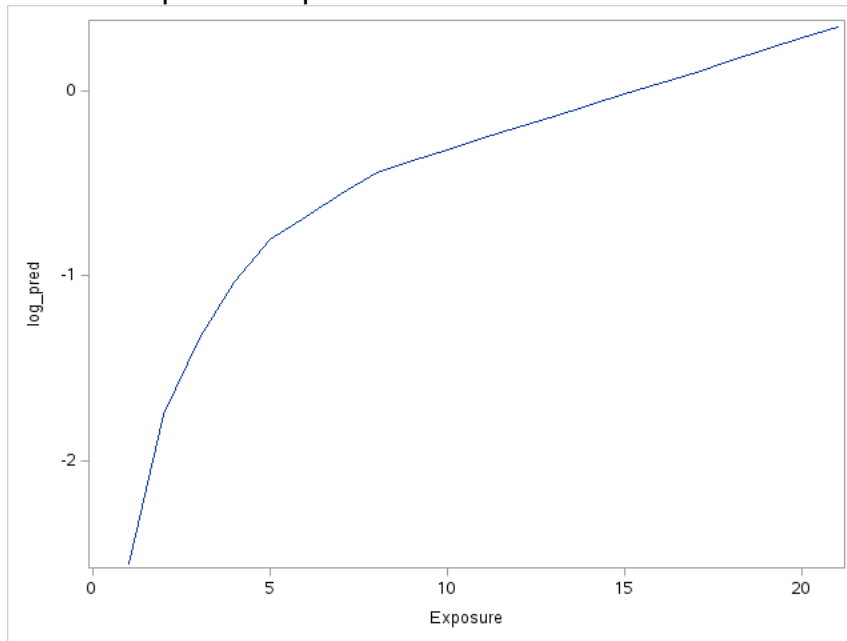


Two things here: a) clearly non-linear, and b) there does appear to be some continuous relationship.

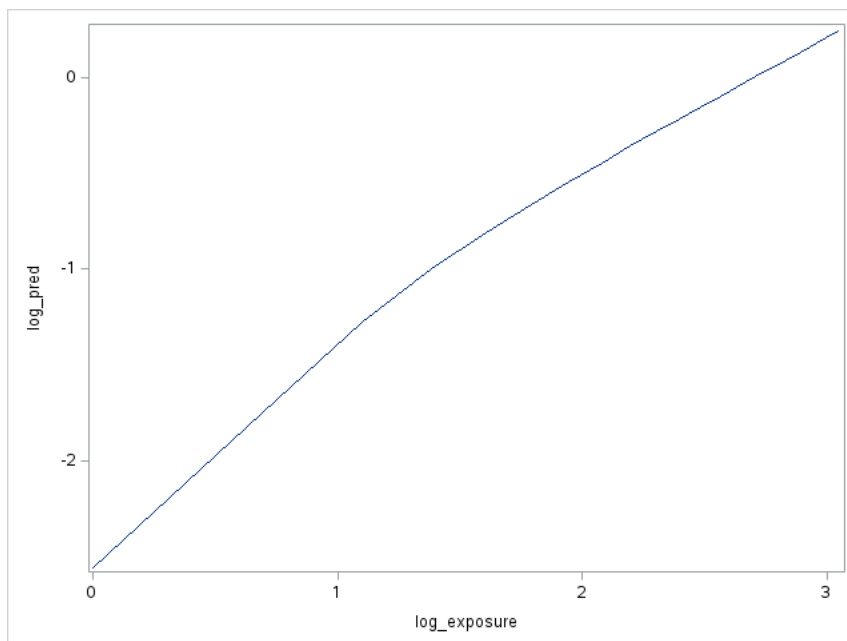
Because of b, I would try to use parametric methods, but I wouldn't blame you for categorizing age here. The downward trend at the end likely isn't there, so we can go with something that will level off.

Really there are two options on the parametric side: splines (which can fit anything), and a cubic age term. I'm going to go with splines because they are always going to fit the data pretty well.

Here's the plot for exposure:



This looks like a classic log-linear trend. Here's what the trend looks like with the log-transformed version of exposure:



This looks pretty linear.

```
proc glimmix data=skin method=quad(QPOINTS=20);  
  class ID Treatment(ref='0') year(ref='1');  
  
  effect age_spl = spline(age / naturalcubic  
    knotmethod=percentiles(6));  
  
  model Y_bin = year Treatment year*Treatment age_spl log_exposure  
    / dist=bin link=logit solution;  
  
  random intercept / subject=ID;  
  
  store out=mod;  
run;
```

Here's the results from the model:

Solutions for Fixed Effects								
Effect	age_spl	Treatment	Year	Estimate	Standard Error	DF	t Value	Pr > t
Intercept				-2.1357	0.7679	7065	-2.78	0.0054
Year			2	-0.04780	0.1541	7065	-0.31	0.7565
Year			3	-0.1078	0.1569	7065	-0.69	0.4920
Year			4	-0.2534	0.1655	7065	-1.53	0.1256
Year			5	0.08608	0.1873	7065	0.46	0.6459
Year			1	0	.	.	.	.
Treatment		1		-0.01229	0.1614	7065	-0.08	0.9393
Treatment		0		0	.	.	.	.
Treatment*Year		1	2	0.09212	0.2135	7065	0.43	0.6661
Treatment*Year		1	3	-0.03075	0.2192	7065	-0.14	0.8884

Solutions for Fixed Effects								
Effect	age_spl	Treatment	Year	Estimate	Standard Error	DF	t Value	Pr > t
Treatment*Year		1	4	0.3104	0.2276	7065	1.36	0.1726
Treatment*Year		1	5	0.1037	0.2628	7065	0.39	0.6931
Treatment*Year		1	1	0	.	.	.	.
Treatment*Year		0	2	0	.	.	.	.
Treatment*Year		0	3	0	.	.	.	.
Treatment*Year		0	4	0	.	.	.	.
Treatment*Year		0	5	0	.	.	.	.
Treatment*Year		0	1	0	.	.	.	.
age_spl	1			0	.	.	.	.
age_spl	2			-0.02270	0.01509	7065	-1.50	0.1327
age_spl	3			0.02114	0.01053	7065	2.01	0.0447
age_spl	4			-0.09273	0.06529	7065	-1.42	0.1556
age_spl	5			0.1252	0.1096	7065	1.14	0.2532
age_spl	6			-0.07186	0.07137	7065	-1.01	0.3141
log_exposure				1.1759	0.06224	7065	18.89	<.0001

Type III Tests of Fixed Effects				
Effect	Num DF	Den DF	F Value	Pr > F
Year	4	7065	1.18	0.3156
Treatment	1	7065	0.70	0.4017

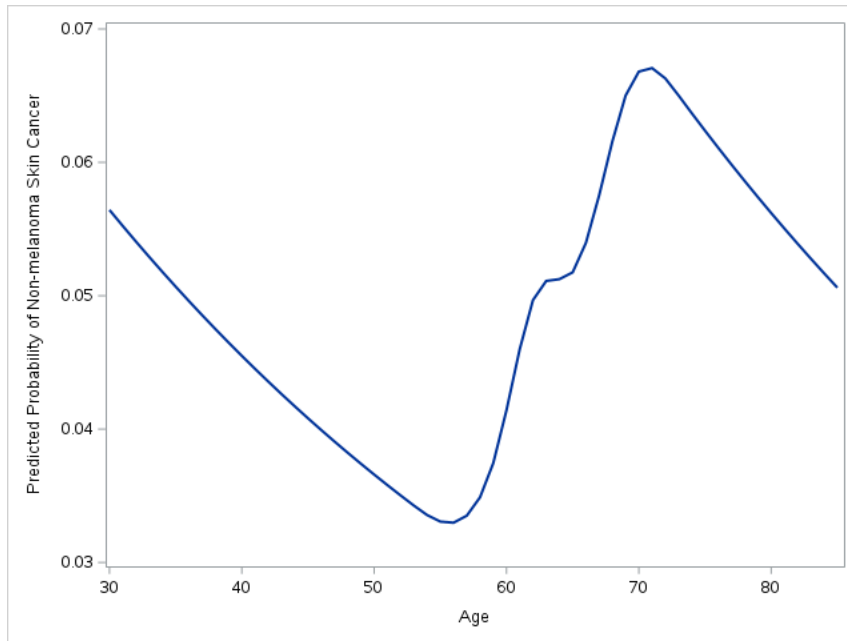
Type III Tests of Fixed Effects				
Effect	Num DF	Den DF	F Value	Pr > F
Treatment*Year	4	7065	0.65	0.6283
age_spl	5	7065	4.77	0.0002
log_exposure	1	7065	356.88	<.0001

Clearly significant age and exposure effects. Here's what the age effect looks like:

```
data age_grid;
  do age = 30 to 85 by 1;
    year = 1;
    Treatment = 0;
    Exposure = mean(Exposure);
    log_exposure = mean(log_exposure);
    output;
  end;
run;

proc plm restore=mod;
  score data=age_grid out=preds / ilink;
run;

proc sgplot data=preds;
  series x=age y=predicted / lineattrs=(thickness=2);
  xaxis label="Age";
  yaxis label="Predicted Probability of Non-melanoma Skin Cancer";
run;
```



Very interesting how different this looks than the original plot. Likely, when we adjusted for the exposure variable we took away (or altered) some of the age effect. It's clearly a non-linear effect (still), but previously it increased till ~ 70 then decreased. Here, It decreases till ~55, then increases till ~ 70, and then decreases thereafter.

The last decrease is likely due to survivorship bias. The people missing data at the end could be due to having (and passing away from) skin cancer or related co-morbidities. That is, the data are missing not at random (MNAR).

(d) (3 points) Does adjusting for exposure and age fit the data better? Give some quantification as to how you made your choice.

The overall conclusion remains the same between the adjusted and unadjusted models (the treatment is not affective), however, the adjusted model fits much better by AIC or BIC.

Unadjusted model:

Fit Statistics	
-2 Log Likelihood	5917.49
AIC (smaller is better)	5939.49
AICC (smaller is better)	5939.53
BIC (smaller is better)	5999.20

Fit Statistics	
CAIC (smaller is better)	6010.20
HQIC (smaller is better)	5961.61

Adjusted model:

Fit Statistics	
-2 Log Likelihood	5488.05
AIC (smaller is better)	5522.05
AICC (smaller is better)	5522.14
BIC (smaller is better)	5614.33
CAIC (smaller is better)	5631.33
HQIC (smaller is better)	5556.23